BITS Pilani

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Introduction to Data Science

Data Wrangling (Contd.), Feature Engineering

- *The slides presented here are obtained from the authors of the books and from various other contributors. I hereby acknowledge all the contributors for their material and inputs.*
- *We have added and modified slides to suit the requirements of the course.*

Handling Numeric Data

Handling Numeric Data

Techniques are

- Discretization – Convert numeric data into discrete categories
- Binarization – Convert numeric data into binary categories
- Normalization – Scale numeric data to a specific range
- Smoothing
  - which may remove noisy variations from the data. Techniques include binning, regression, and clustering.

Pima Indians Diabetes dataset

- Numeric attributes com with diverse ranges.

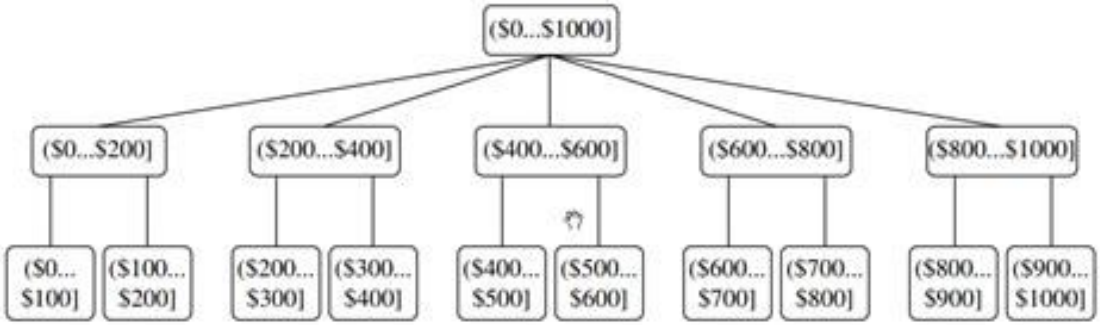
Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesF	Age	Outcome
6	148	72	35	0	34	0.627	50	1
1	85	66	29	0	27	0.351	31	0
8	183	64	0	0	23	0.672	32	1
1	89	66	23	94	28	0.167	21	0
0	137	40	35	168	43	2.288	33	1
5	116	74	0	0	26	0.201	30	0
3	78	50	32	88	31	0.248	26	1
10	115	0	0	0	35	0.134	29	0
2	197	70	45	543	31	0.158	53	1
8	125	96	0	0	0	0.232	54	1
4	110	92	0	0	38	0.191	30	0
10	168	74	0	0	38	0.537	34	1
10	139	80	0	0	27	1.441	57	0
1	189	60	23	846	30	0.398	59	1

Discretization

- Convert continuous attribute into a discrete attribute.
- Discretization involves converting the raw values of a numeric attribute (e.g., age) into
  - interval labels (e.g., 0–10, 11–20, etc.)
  - conceptual labels (e.g., youth, adult, senior)
- Discretization Process
  - The raw data are replaced by a smaller number of interval or concept labels.
  - This simplifies the original data and makes the mining more efficient.
  - Concept hierarchies are also useful for mining at multiple abstraction levels.

Concept Hierarchy

- Divide the range of a continuous attribute into intervals.
- Interval labels can then be used to replace actual data values.
- The labels, in turn, can be recursively organized into higher-level concepts.
- This results in a concept hierarchy for the numeric attribute.



A concept hierarchy for the attribute *price*, where an interval ($X . . . $Y] denotes the range from $X (exclusive) to $Y (inclusive).

Discretization Techniques

Discretization techniques can be categorized based on how the discretization is performed.

- Supervised vs. Unsupervised discretization
  - If the discretization process uses class information, then we say it is supervised discretization. Otherwise, it is unsupervised.
- Top-down discretization or Splitting
  - The process starts by first finding one or a few points (called split points or cut points) to split the entire attribute range. Then the process repeats recursively on the resulting intervals.
- Bottom-up discretization or Merging
  - The process starts by considering all of the continuous values as potential split-points.
    - Removes some by merging neighborhood values to form intervals. Then recursively applies this process to the resulting intervals.

Discretization by Binning Methods

1. Equal Width (distance) binning
   - Each bin has equal width.

$$width = interval = \frac{\max - \min}{\#bins}$$

   - Highly sensitive to outliers.
   - If outliers are present, the width of each bin is large, resulting in skewed data.
2. Equal Depth (frequency) binning
   - Specify the number of values that have to be stored in each bin.
   - Number of entries in each bin are equal.
   - Some values can be stored in different bins.

Binning Example

Discretize the following data into 3 discrete categories using binning technique.

70, 70, 72, 73, 75, 75, 76, 76, 78, 79, 80, 81, 53, 56, 57, 63, 66, 67, 67, 67, 68, 69, 70, 70.

Binning Example

Original Data	53, 56, 57, 63, 66, 67, 67, 67, 68, 69, 70, 70, 70, 70, 72, 73, 75, 75, 76, 76, 78, 79, 80, 81			
Method		Bin1	Bin 2	Bin 3
Equal	width=	[53, 62) =	[62, 72) =	[72, 81] =
Width	81-53 = 28	53, 56, 57	63, 66, 67, 67,	72, 73, 75, 75,
	28/3 = 9.33		67, 68, 69, 70,	76, 76, 78,
			70, 70, 70	79, 80, 81
Equal	depth =	53, 56, 57, 63,	68, 69, 70, 70,	75, 75, 76, 76,
Depth	24 /3 = 8	66, 67, 67, 67	70, 70, 72, 73	78, 79, 80, 81

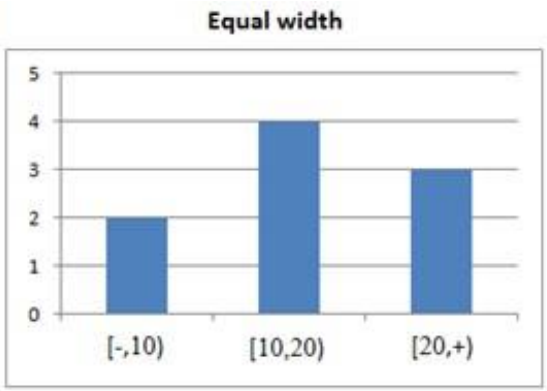
Discretization by Histogram Analysis

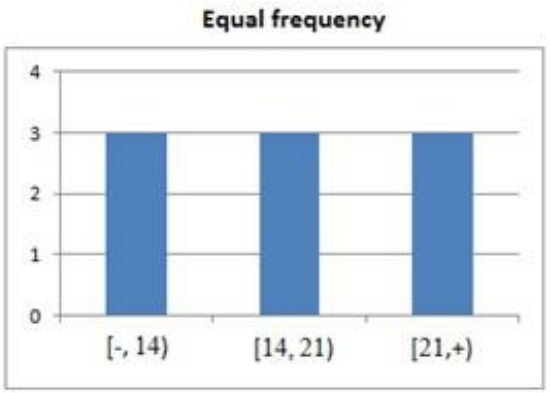
- Histogram analysis is an unsupervised discretization technique because it does not use class information.
- Histograms use binning to approximate data distributions and are a popular form of data reduction.
- A histogram for an attribute, X, partitions the data distribution of X into disjoint subsets, referred to as buckets or bins.
- If each bucket represents only a single attribute–value/frequency pair, the buckets are called singleton buckets.
- Often, buckets represent continuous ranges for the given attribute.
- The histogram analysis algorithm can be applied recursively to each partition in order to automatically generate a multilevel concept hierarchy.

Discretization by Histogram Analysis

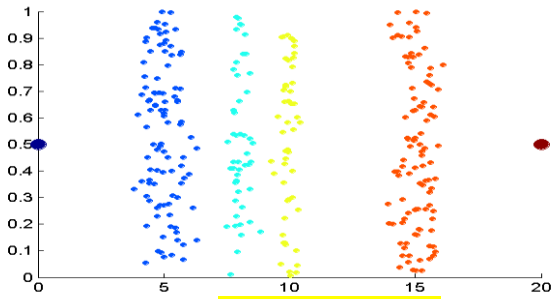
1. Equal Width Histogram
   - The values are partitioned into equal size partitions or ranges.
2. Equal Frequency Histogram
   - The values are partitioned such that each partition contains the same number of data objects.

- **Data :** 0, 4, 12, 16, 16, 18, 24, 26, 28
- **Equal width**
  - Bin 1: 0, 4 [-,10)
  - Bin 2: 12, 16, 16, 18 [10,20)
  - Bin 3: 24, 26, 28 [20,+)
- **Equal frequency**
  - Bin 1: 0, 4, 12 [-, 14)
  - Bin 2: 16, 16, 18 [14, 21)
  - Bin 3: 24, 26, 28 [21,+)

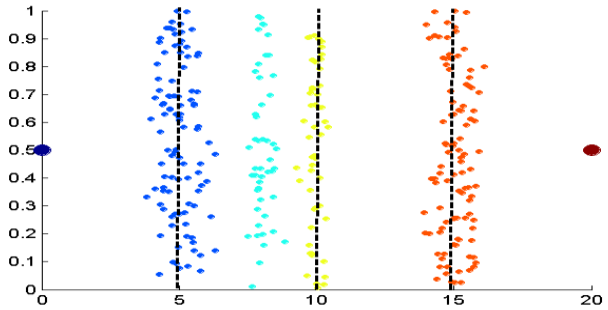




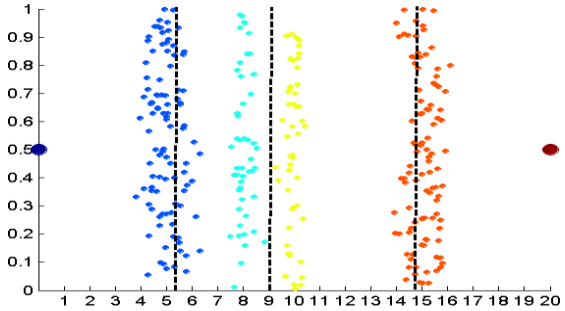
Discretization Without Supervision: Binning vs. Clustering



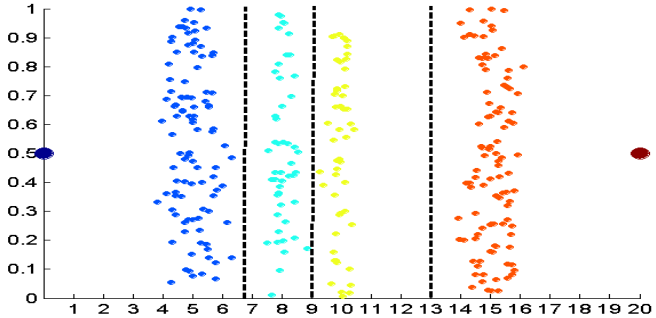
Data



Equal width (distance) binning



Equal depth (frequency) (binning)



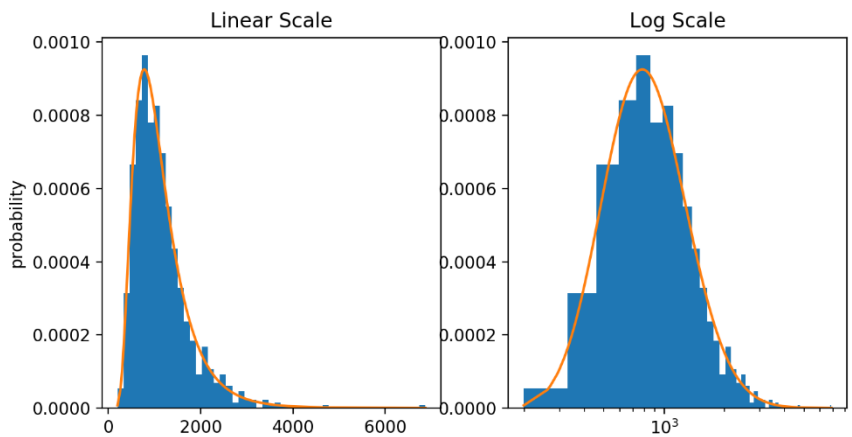
K-means clustering leads to better results

Variable Transformation

- Variable transformation involves changing the values of an attribute.
- For each object (tuple), a transformation is applied to the value of the variable for that object.
  1. Simple functional transformations
  2. Normalization

Simple Functional Transformation

- For this type of variable transformation, a simple mathematical function is applied to each value individually.
- For a variable x, simple transformations include
  - x^k, $\log x$, e^x, $\sqrt{x}$, $\frac{1}{x}$, $\sin x$, $|\ x\ |$
- In statistics, variable transformations, especially $\log x$, $\sqrt{x}$, $\frac{1}{x}$, are often used to transform data that does not have a Gaussian (normal) distribution into data that does. While this can be important, other reasons often take precedence in data mining.
- Eg: Transfer of data bytes may be represented in the the $\log_{10}$ transformation.



https://aegis4048.github.io/transforming-non-normal-distribution-to-normal-distribution

Normalization

- Normalizing the data attempts to give all attributes an equal weight.
- The goal of standardization or normalization is to make an entire set of values have a particular property.
- Normalization is particularly useful for:
  - classification algorithms involving neural networks.
    - normalizing the input values for each attribute in the training tuples will help speed up the learning phase.
  - distance measurements such as nearest-neighbor classification and clustering.
    - normalization helps prevent attributes with initially large ranges (e.g., income) from outweighing attributes with initially smaller ranges (e.g., binary attributes).

Why Feature Scaling?

- Features with bigger magnitude dominate over the features with smaller magnitudes.
- Good practice to have all variables within a similar scale.
- Euclidean distances are sensitive to feature magnitude.
- Gradient descent converges faster when all the variables are in the similar scale.
- Feature scaling helps decrease the time of finding support vectors.

Why Feature Scaling?

- For distance-based methods, normalization helps prevent attributes with initially large ranges (e.g., income) from out-weighing attributes with initially smaller ranges (e.g., binary attributes).

Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesF	Age	Outcome
6	148	72	35	0	34	0.627	50	1
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10	168	74	0	0	38	0.537	34	1
10	139	80	0	0	27	1.441	57	0
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Algorithms Sensitive to Feature Magnitude

- Linear and Logistic Regression
- Neural Networks
- Support Vector Machines
- KNN
- K-Means Clustering
- Linear Discriminant Analysis (LDA)
- Principal Component Analysis (PCA)

Normalization

- Scale the feature magnitude to a standard range like [0, 1] or [−1, +1] or any other.
- Techniques
  - Min-Max normalization
  - z-score normalization
  - Normalization by decimal scaling
- Impact of outliers in the data ???

Min-Max Scaling

- Min-max scaling squeezes (or stretches) all feature values to be within the range of [0, 1].
- Min-Max normalization preserves the relationships among the original data values.
- It will encounter an ”out-of-bounds” error if a future input case for normalization falls outside of the original data range for X .

$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)} \qquad \text{for range}[0, 1]$$

$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)}(new_{max} - new_{min}) + new_{min} \qquad \text{for range}[new_{min}, new_{max}]$$

Min-Max Normalization

Suppose that the minimum and maximum values for the attribute income are \$12,000 and \$98,000, respectively. The new range is [0.0,1.0]. Apply min-max normalization to value of \$73,600.

$$\hat{x} = \frac{x - \min(x)}{\max(x) - \min(x)}(new_{max} - new_{min}) + new_{min}$$
$$= \frac{73600 - 12000}{98000 - 12000}(1.0 - 0.0) + 0.0$$
$$= 0.716$$

z-score Normalization

- In z-score normalization (or zero-mean normalization), the values for an attribute, x , are normalized based on the mean $\mu(x)$ and standard deviation $\sigma(x)$ of x .
- The resulting scaled feature has a mean of 0 and a variance of 1.
- Z score is

$$\frac{x - \mu(x)}{\sigma(x)}$$

- z-score normalization is useful when the actual minimum and maximum of attribute X are unknown, or when there are outliers that dominate the min-max normalization.

z-score Normalization

Suppose that the mean and standard deviation of the values for the attribute income are $54,000 and $16,000, respectively. Apply z-score normalization to value of $73,600.

$$\begin{aligned}\hat{x} &= \frac{x - \mu(x)}{\sigma(x)} \\ &= \frac{73600 - 54000}{16000} \\ &= 1.225\end{aligned}$$

Decimal Normalization

- Normalizes by moving the decimal point of values of attribute x .
- The number of decimal points moved depends on the maximum absolute value of x .
- New range is $[-1, +1]$.

$$j = \text{smallest integer such that } \max(|\hat{x}|) < 1$$

$$\hat{x} = \frac{x}{10^j}$$

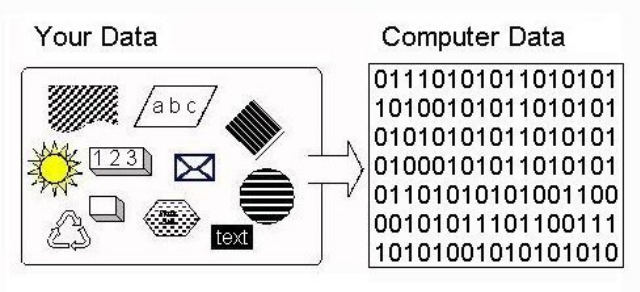
Decimal Normalization

Example 1		
CGPA	Formula	Normalized CGPA
2	2/10	0.2
3	3/10	0.3
Example 2		
Bonus	Formula	Normalized Bonus
450	450/1000	0.45
310	310/1000	0.31
Example 3		
Salary	Formula	Normalized Salary
48000	48000/100000	0.48
67000	67000/100000	0.67

CATEGORICAL ENCODING

- We need to convert categorical columns to numeric columns so that a machine learning algorithm understands it.
- Categorical encoding is a process of converting categories to numbers.

- Binarization maps a continuous or categorical attribute into one or more binary attributes.
- Must maintain ordinal relationship.
- Algorithms that find association patterns require that the data be in the form of binary attributes.
- E.g., Apriori algorithm, Frequent Pattern (FP) Growth algorithm



CATEGORICAL ENCODING Techniques

- One-hot encoding
- Label Encoding

One-Hot Encoding

- Encode each categorical variable with a set of Boolean variables which take values 0 or 1, indicating if a category is present for each observation.
- One binary attribute for each categorical value.
- Advantages
  - Makes no assumption about the distribution or categories of the categorical variable .
  - Keeps all the information of the categorical variable .
  - Suitable for linear models.
- Disadvantages
  - Expands the feature space.
  - Does not add extra information while encoding.
  - Many dummy variables may be identical, introducing redundant information .
  - Number of resulting attributes may become too large.
- In multi-class classification, the class label is converted using one-hot encoding.

One-Hot Encoding Example

- Assume an ordinal attribute for representing service of a restaurant: (*Awful* < *Poor* < *OK* < *Good* < *Great*) requires 5 bits to maintain the ordinal relationship.

Service Quality	X1	X2	X3	X4	X5
Awful	0	0	0	0	1
Poor	0	0	0	1	0
OK	0	0	1	0	0
Good	0	1	0	0	0
Great	1	0	0	0	0

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One-hot Encoding Example

id	color
1	red
2	blue
3	green
4	blue

One Hot Encoding

id	color_red	color_blue	color_green
1	1	0	0
2	0	1	0
3	0	0	1
4	0	1	0

Label Encoding

- Replace the categories by digits from 1 to *n* (or 0 to *n* − 1, depending the implementation), where *n* is the number of distinct categories of the variable.
- The categories are arranged in ascending order and the numbers are assigned.
- Advantages
  - Straightforward to implement.
  - Does not expand the feature space.
  - Work well enough with tree based algorithms.
- Disadvantages
  - Does not add extra information while encoding.
  - Not suitable for linear models.
  - Does not handle new categories in test set automatically.
- Used for features which have multiple values into domain. eg: colour, protocol types

Label Encoding Example

- Assume an ordinal attribute for representing service of a restaurant: (Awful, Poor, OK, Good, Great)

Service Quality	Integer Value
Awful	0
Poor	1
OK	2
Good	3
Great	4

Example for Proximity

Example

Based on the information given in the table below, find most similar and most dissimilar persons among them. Apply min-max normalization on income to obtain [0,1] range. Consider profession and mother tongue as nominal. Consider native place as ordinal variable with ranking order of [Village, Small Town, Suburban, Metropolitan]. Give equal weight to each attribute.

Name	Income	Profession	Mother tongue	Native Place
Ram	70000	Doctor	Bengali	Village
Balram	50000	Data Scientist	Hindi	Small Town
Bharat	60000	Carpenter	Hindi	Suburban
Kishan	80000	Doctor	Bhojpuri	Metropolitan

Solution

After normalizing income and quantifying native place, we get

Name	Income	Profession	Mother tongue	Native Place
Ram	0.67	Doctor	Bengali	1
Balram	0	Data Scientist	Hindi	2
Bharat	0.33	Carpenter	Hindi	3
Kishan	1	Doctor	Bhojpuri	4

d(Ram, Balram) = 0.67+1+1+(2-1)/(4-1)=3 d(Ram, Bharat) = 0.33+1+1+(3-1)/(4-1)=3

d(Ram, Kishan) = 0.33+0+1+(4-1)/(4-1) = 2.33 d(Balram, Bharat) = 0.33+1+0+(3-2)/(4-1)=1.67

d(Balram, Kishan) = 1+1+1+(4-2)/(4-1) = 3.67 d(Bharat, Kishan) = 0.67+1+1+(4-3)/(4-1) = 3

Most similar – Balram and Bharat; Most dissimilar – Balram and Kishan

Feature Engineering

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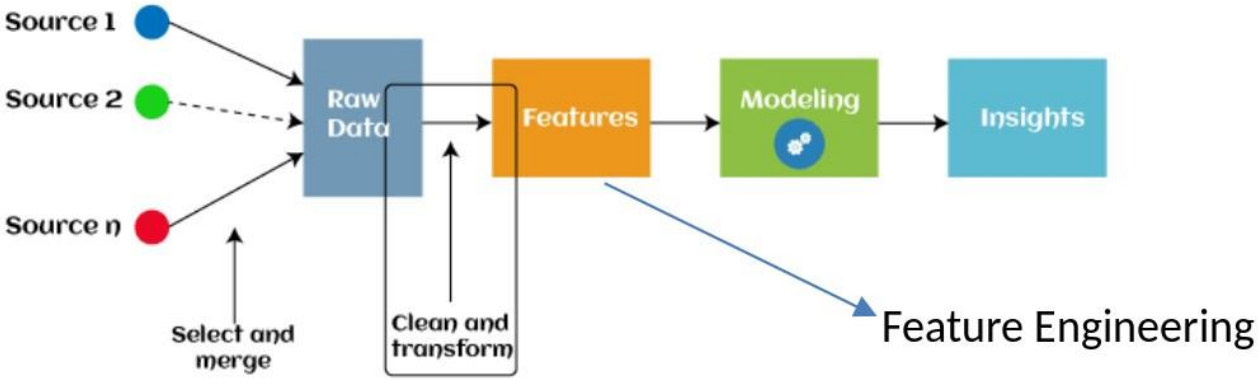
Feature Engineering

- Feature Engineering is the process of selecting and extracting useful, predictive features from data.
- The goal is to create a set of features that best represent the information contained in the data, producing a simpler model that generalizes well to future observations.



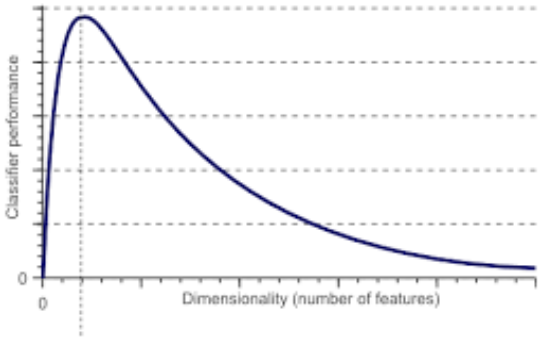
Motivation for Feature Engineering

HUGHES PHENOMENON

Given fixed number of data points, performance of a regressor or a classifier first increases but later decreases as the number of dimensions of the data increases.

Reasons for this phenomenon

- Redundant Features
- Correlation between features
- Irrelevant Features



Handling Redundancy

Handling Redundancy in Data Integration

- Redundant data occur often when integration of multiple databases
  - *Object identification*: The same attribute or object may have different names in different databases
  - *Derivable data:* One attribute may be a "derived" attribute in another table, e.g., annual revenue
- Redundant attributes may be able to be detected by *correlation analysis* and *covariance analysis*
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Correlation Analysis (Numeric Data)

- Correlation coefficient (also called Pearson's product moment coefficient)

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \overline{A})(b_i - \overline{B})}{n\sigma_A\sigma_B} = \frac{\sum_{i=1}^{n}(a_i b_i) - n\overline{AB}}{n\sigma_A\sigma_B}$$

where n is the number of tuples, $\overline{A}$ and $\overline{B}$ are the respective means of A and B, σ_A and σ_B are the respective standard deviation of A and B.

- $+1 >= r_{A,B} >= -1$
- If $r_{A,B} > 0$, A and B are positively correlated (A's values increase as B's). The higher, the stronger correlation.
- $r_{A,B} = 0$: independent;
- $r_{A,B} < 0$: negatively correlated

Correlation (viewed as linear relationship)

- Correlation measures the linear relationship between objects
- To compute correlation, we standardize data objects, A and B, and then take their dot product

$$a'_k = (a_k - mean(A)) / std(A)$$

$$b'_k = (b_k - mean(B)) / std(B)$$

$$correlation(A, B) = A' \bullet B'$$

Covariance (Numeric Data)

- Covariance is similar to correlation

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

Correlation coefficient: $$r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$$

where n is the number of tuples, $\bar{A}$ and $\bar{B}$ are the respective mean or **expected values** of A and B, σ_A and σ_B are the respective standard deviation of A and B

- **Positive covariance**: If $Cov_{A,B} > 0$, then A and B both tend to be larger than their expected values
- **Negative covariance**: If $Cov_{A,B} < 0$ then if A is larger than its expected value, B is likely to be smaller than its expected value
- **Independence**: $Cov_{A,B} = 0$

Co-Variance: An Example

- It can be simplified in computation as

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

- Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14).

$$Cov(A,B) = E(A \cdot B) - \bar{A}\bar{B}$$

- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
  - E(A) = (2 + 3 + 5 + 4 + 6)/ 5 = 20/5 = 4
  - E(B) = (5 + 8 + 10 + 11 + 14) /5 = 48/5 = 9.6
  - Cov(A,B) = (2×5+3×8+5×10+4×11+6×14)/5 − 4 × 9.6 = 4
- Thus, A and B rise together since Cov(A, B) > 0.

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Correlation Analysis (Nominal Data)

- **χ^2 (chi-square) test**

$$\chi^2 = \sum \frac{(Observed - Expected)^2}{Expected}$$

- The larger the **χ^2** (chi-square) value, the more likely the variables are related
- The cells that contribute the most to the **χ^2** value are those whose actual count is very different from the expected count
- Correlation does not imply causality
  - # of hospitals and # of car-theft in a city are correlated
  - Both are causally linked to the third variable: population

Chi-Square Calculation: An Example

	Play chess	Not play chess	Sum (row)
Like science fiction	250	200	450
Not like science fiction	50	1000	1050
Sum(col.)	300	1200	1500

Do you think there is correlation between 'Play chess' and 'Like science fiction'

For the above cross-tabulation

- There are two possible values in rows: Likes science fiction, Not like science fiction
- There are two possible values in columns: Play chess, Not play chess

Degrees of Freedom = (rows-1)*(cols-1) = (2-1)(2-1) = 1

Chi-Square Calculation: Computation

	Play chess	Not play chess	Sum (row)
Like science fiction	250(90)	200(360)	450
Not like science fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

- X^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)

$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$$

- We can check statistical significance of chi-square value in standard (χ^2) table.
- It shows that like_science_fiction and play_chess are correlated in the dataset

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	P										
DF	0.995	0.975	0.2	0.1	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	.0004	.00016	1.642	2.706	3.841	5.024	5.412	6.635	7.879	9.55	10.828
2	0.01	0.0506	3.219	4.605	5.991	7.378	7.824	9.21	10.597	12.429	13.816
3	0.0717	0.216	4.642	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	0.207	0.484	5.989	7.779	9.488	11.143	11.668	13.277	14.86	16.924	18.467
5	0.412	0.831	7.289	9.236	11.07	12.833	13.388	15.086	16.75	18.907	20.515
6	0.676	1.237	8.558	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458
7	0.989	1.69	9.803	12.017	14.067	16.013	16.622	18.475	20.278	22.601	24.322
8	1.344	2.18	11.03	13.362	15.507	17.535	18.168	20.09	21.955	24.352	26.124
9	1.735	2.7	12.242	14.684	16.919	19.023	19.679	21.666	23.589	26.056	27.877
10	2.156	3.247	13.442	15.987	18.307	20.483	21.161	23.209	25.188	27.722	29.588
11	2.603	3.816	14.631	17.275	19.675	21.92	22.618	24.725	26.757	29.354	31.264
12	3.074	4.404	15.812	18.549	21.026	23.337	24.054	26.217	28.3	30.957	32.909
13	3.565	5.009	16.985	19.812	22.362	24.736	25.472	27.688	29.819	32.535	34.528
14	4.075	5.629	18.151	21.064	23.685	26.119	26.873	29.141	31.319	34.091	36.123
15	4.601	6.262	19.311	22.307	24.996	27.488	28.259	30.578	32.801	35.628	37.697
16	5.142	6.908	20.465	23.542	26.296	28.845	29.633	32	34.267	37.146	39.252
17	5.697	7.564	21.615	24.769	27.587	30.191	30.995	33.409	35.718	38.648	40.79
18	6.265	8.231	22.76	25.989	28.869	31.526	32.346	34.805	37.156	40.136	42.312
19	6.844	8.907	23.9	27.204	30.144	32.852	33.687	36.191	38.582	41.61	43.82
20	7.434	9.591	25.038	28.412	31.41	34.17	35.02	37.566	39.997	43.072	45.315

Feature Construction

Feature Construction

- Create derived features
- Involves creating a new feature using data from existing features
- Mostly rely on domain knowledge
- Eg: Calculating price per sqft

Area	Price (Rs)	Price/Sft (Rs)
1800	81,00,000	4500
2000	78,00,000	3900
1550	65,10,000	4200
2400	1,15,20,000	4800
3500	1,22,50,000	3500
2800	1,45,60,000	5200

Feature Construction

Customer ID	Gender	Session Begin	Session End
C001	Female	15-06-2019 10:30	15-06-2019 11:15
C002	Male	13-06-2019 08:00	13-06-2019 08:03
C003	Female	02-06-2019 16:25	02-06-2019 18:35
C004	Male	01-06-2019 11:20	01-06-2019 13:00

Customer ID	Gender	Session Duration
C001	Female	45
C002	Male	3
C003	Female	125
C004	Male	100

Feature Selection

BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

Feature Selection Methods

- Forward selection
  - starts with one predictor and adds more iteratively.
  - At each subsequent iteration, the best of the remaining original predictors are added based on performance criteria.
  - *SequentialFeatureSelector class from mlxtend*
- Backward elimination
  - starts with all predictors and eliminates one-by-one iteratively.
  - One of the most popular algorithms is Recursive Feature Elimination (RFE) which eliminates less important predictors based on feature importance ranking.
  - *RFE class from sklearn*

SFS Example – wine data

	feature_idx	cv_scores	avg_score	feature_names
1	(6,)	[0.8426966292134831]	0.842697	(flavanoids,)
2	(6, 9)	[0.9438202247191011]	0.94382	(flavanoids, color_intensity)
3	(6, 8, 9)	[0.949438202247191]	0.949438	(flavanoids, proanthocyanins, color_intensity)
4	(0, 6, 8, 9)	[0.9662921348314607]	0.966292	(alcohol, flavanoids, proanthocyanins, color_i...
5	(0, 5, 6, 8, 9)	[0.9775280898876404]	0.977528	(alcohol, total_phenols, flavanoids, proanthoc...
6	(0, 5, 6, 7, 8, 9)	[0.9719101123595506]	0.97191	(alcohol, total_phenols, flavanoids, nonflavan...
7	(0, 2, 5, 6, 7, 8, 9)	[0.9662921348314607]	0.966292	(alcohol, ash, total_phenols, flavanoids, nonf...

SBS Example – wine data

	feature_idx	cv_scores	avg_score	feature_names
13	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12)	[0.8258426966292135]	0.825843	(alcohol, malic_acid, ash, alcalinity_of_ash, ...
12	(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)	[0.9269662921348315]	0.926966	(alcohol, malic_acid, ash, alcalinity_of_ash, ...
11	(0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11)	[0.9550561797752809]	0.955056	(alcohol, malic_acid, ash, alcalinity_of_ash, ...
10	(0, 2, 3, 5, 6, 7, 8, 9, 10, 11)	[0.9662921348314607]	0.966292	(alcohol, ash, alcalinity_of_ash, total_phenol...
9	(0, 2, 3, 6, 7, 8, 9, 10, 11)	[0.9719101123595506]	0.97191	(alcohol, ash, alcalinity_of_ash, flavanoids, ...
8	(0, 2, 3, 6, 7, 8, 10, 11)	[0.9719101123595506]	0.97191	(alcohol, ash, alcalinity_of_ash, flavanoids, ...
7	(0, 2, 3, 6, 7, 8, 11)	[0.9775280898876404]	0.977528	(alcohol, ash, alcalinity_of_ash, flavanoids, ...
6	(0, 2, 3, 6, 8, 11)	[0.9775280898876404]	0.977528	(alcohol, ash, alcalinity_of_ash, flavanoids, ...

Text Books

T1	Introduction to Data Mining, by Tan, Steinbach and Vipin Kumar
T2	Introducing Data Science by Cielen, Meysman and Ali
T3	Storytelling with Data, A data visualization guide for business professionals, by Cole Nussbaumer Knaflic; Wiley
T4	Data Mining: Concepts and Techniques, Third Edition by Jiawei Han and Micheline Kamber Morgan Kaufmann Publishers

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Thank You